

Advanced Java

BRANCH: INFORMATION TECH (IT)

COURSE CODE: IT304

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Advanced Java (IT304) under the Information Tech (IT) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Multithreading & Collection Framework

Java Thread lifecycle, Synchronization, Runnable interface, thread pools. Collections: List, Set, Map, ArrayList, HashMap, Iterator.

Unit 2: Java Web Development

Servlet lifecycle, request and response handling, session tracking (Cookies, HttpSession). JSP (Java Server Pages): scripting elements, JSP directives.

Unit 3: Database & Enterprise Java

JDBC architecture, drivers, executing SQL queries, PreparedStatement. Enterprise Java Beans (EJB) intro, RMI.

Unit 4: Spring Framework

Dependency Injection (DI), Inversion of Control (IoC) container. Spring MVC architecture. Spring Boot: auto-configuration, building RESTful Web Services.

Servlet Methods: init(), service(), destroy(). JDBC connection: DriverManager.getConnection(url, user, pwd).

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



